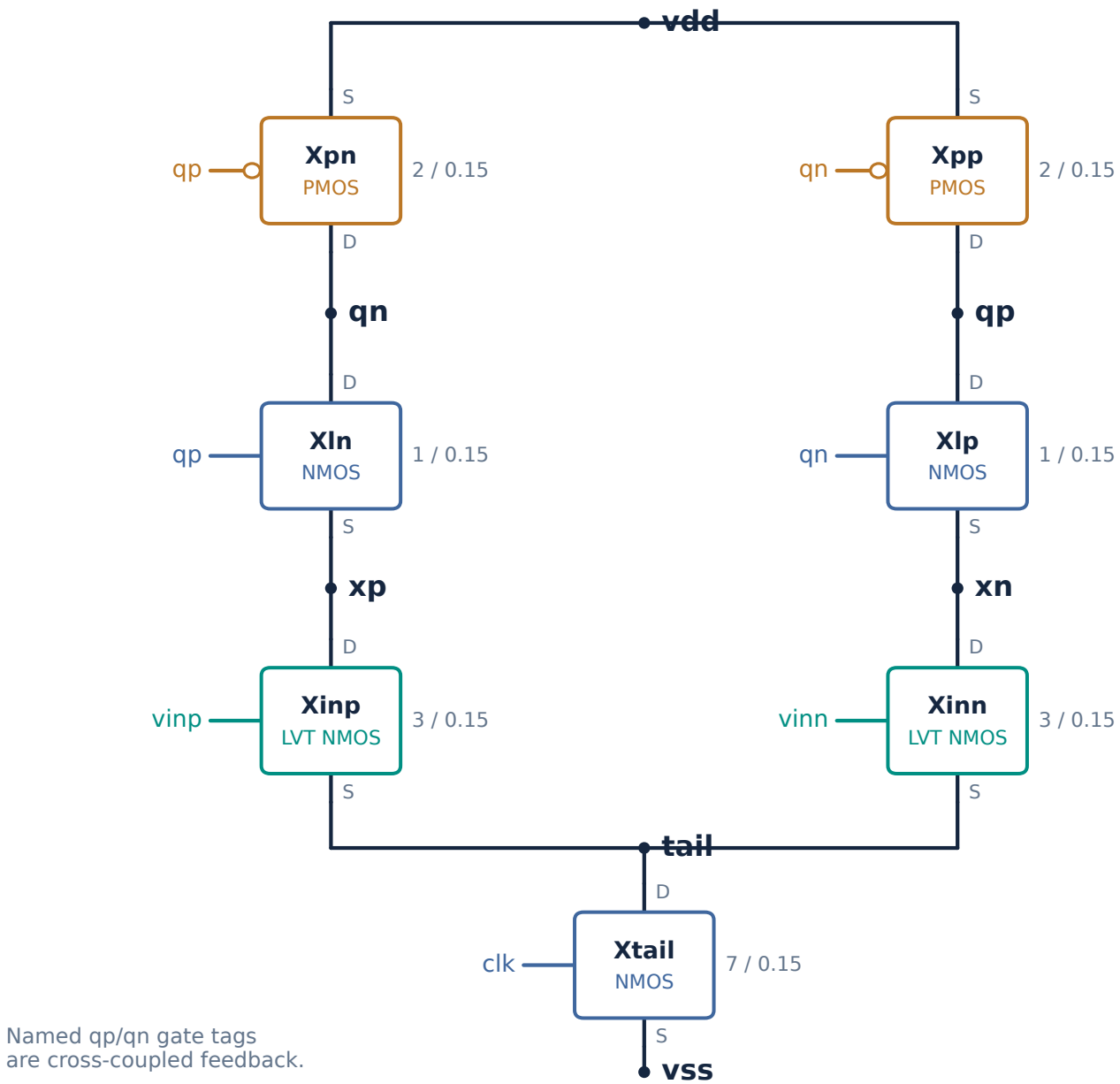


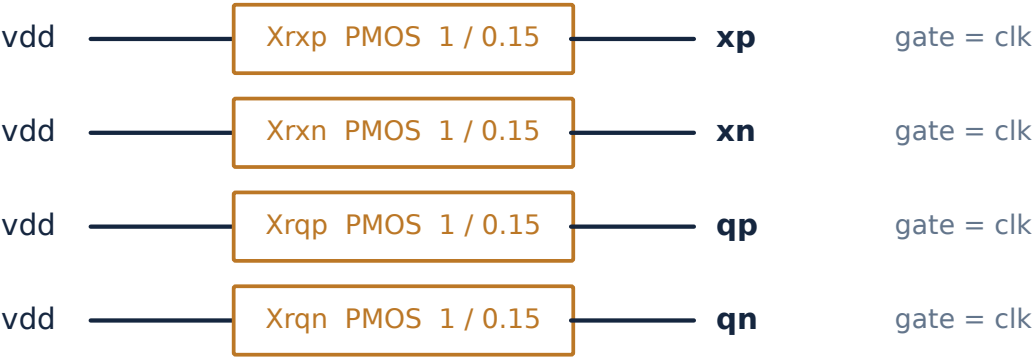
Comparator Atlas | Circuit guide

Published 27-transistor design: conventional StrongARM core + physical source-gated trim

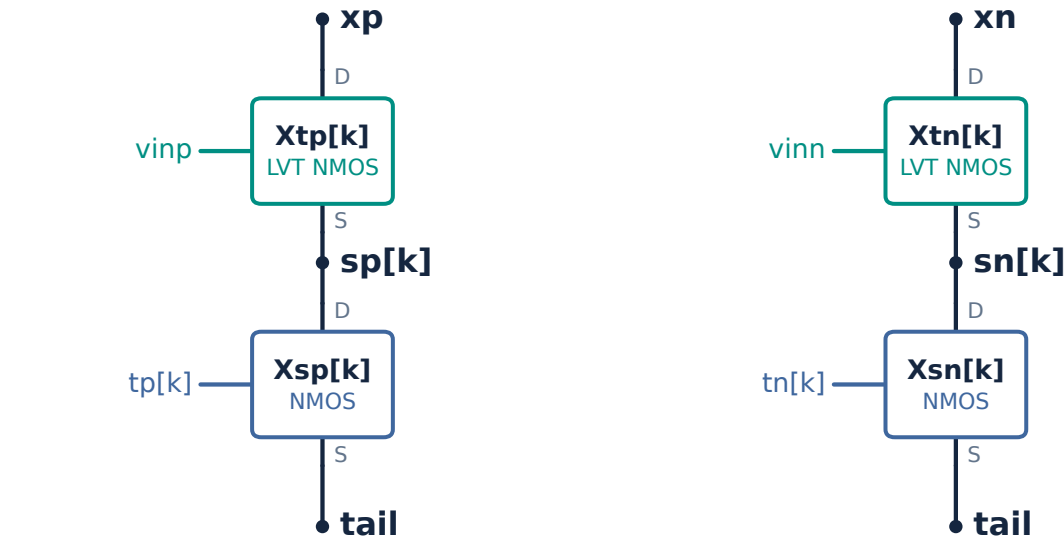
A. Regenerative core (7 devices)



B. Clocked reset devices (4 devices)



C. Expand one trim bit on each side (16 devices total)



Repeat $k = 0, 1, 2, 3$ on BOTH sides. No ideal offset subtraction.

Magnitude bits k	0	1	2	3
Auxiliary LVT W/L	.42/1	.84/1	1.68/1	3.36/1
Source switch W/L	1/.15	2/.15	4/.15	8/.15

Polarity

$v_{inp} > v_{inn}$ discharges x_p / q_n first; a correct positive decision has q_p high and q_n low.

All W/L values are nominal micrometres. Every NMOS bulk = vss; every PMOS bulk = vdd (bulk wires omitted).

Code 0 keeps all trim devices physically present. Sign + four magnitude bits select codes -15 through +15.

Connectivity guide, not GDS or a performance claim. External 5 fF loads, ideal source drivers and controller are not drawn.